

Data Wrangling in R

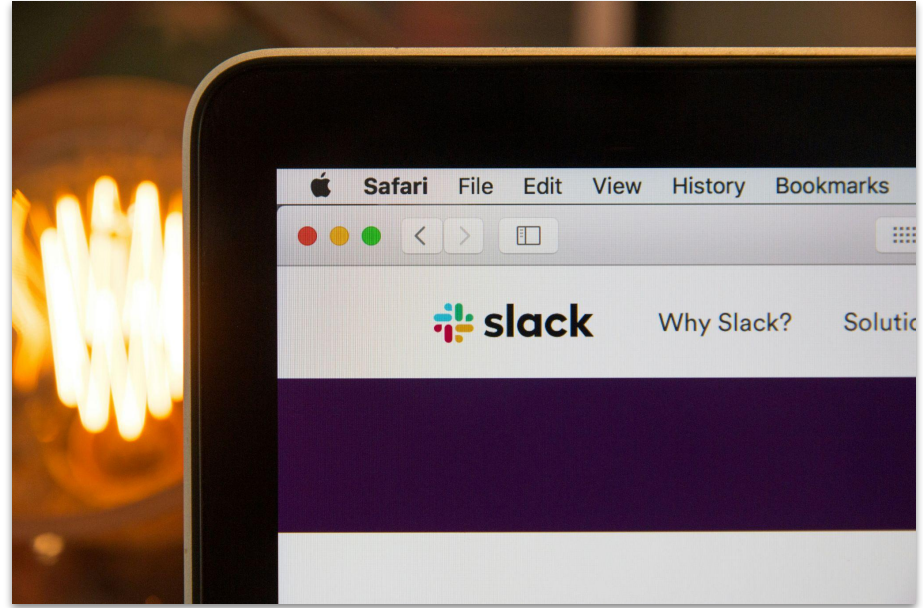
<http://sisbid.github.io/Data-Wrangling/>

Course Info

Course name	Data Wrangling in R
Instructors	<u>Carrie Wright</u> and <u>Ava Hoffman</u>
Course website	<u>http://sisbid.github.io/Data-Wrangling/</u>
Goals	Teach you how to import and clean data
Pre-reqs	Hopefully some R programming

Slack is helpful for:

- Getting help
- Talking with peers
- Accessing recordings



What we will cover in this course:

- 1) The importance of data wrangling
- 2) Importing data (and outporting)
- 3) Subsetting data
- 4) Summarizing data
- 5) Cleaning data
- 6) Reshaping data
- 7) Data merging and joining
- 8) Functional programming (efficiently applying functions)
- 9) Working with factors, strings, dates
- 10) Reproducibility practices like version control - Git/GitHub

But first, some jargon!

Packages

A bundle or “package” of code (and or possibly data) that can be loaded together for easy repeated use or for sharing with others.

Packages are analogous to a software application like Microsoft Word on your computer.



Function

A piece of code that allows you to do something in R. Packages often contain functions.

You can think of a function as verb in R.

A function might help you add numbers together, create a plot, or organize your data. More on that soon!

```
sum(1, 20234)
```

```
[1] 20235
```

Argument

Something you give to a function



```
round(0.627, digits = 2)
```

```
[1] 0.63
```

An Object

Something that can be worked with or on in R - can be lots of different things!

You can think of an objects as a noun in R.

An object might be a data table, a plot, a function, or more!

Tree	age	circumference
1	118	30
1	484	58
1	664	87
1	1004	115
1	1231	120
1	1372	142
1	1582	145
2	118	33

Dataframes/DataTables/Spreadsheets

		Rows			
Columns					

Dataframes/Data tables

Rows = samples - individuals, locations, houses, viruses etc.

Columns = variables - aspects or features measured, color, count, type etc.

```
head(iris)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

Need more help?

R jargon:

<https://link.springer.com/content/pdf/bbm%3A978-1-4419-1318-0%2F1.pdf>

The Tidyverse

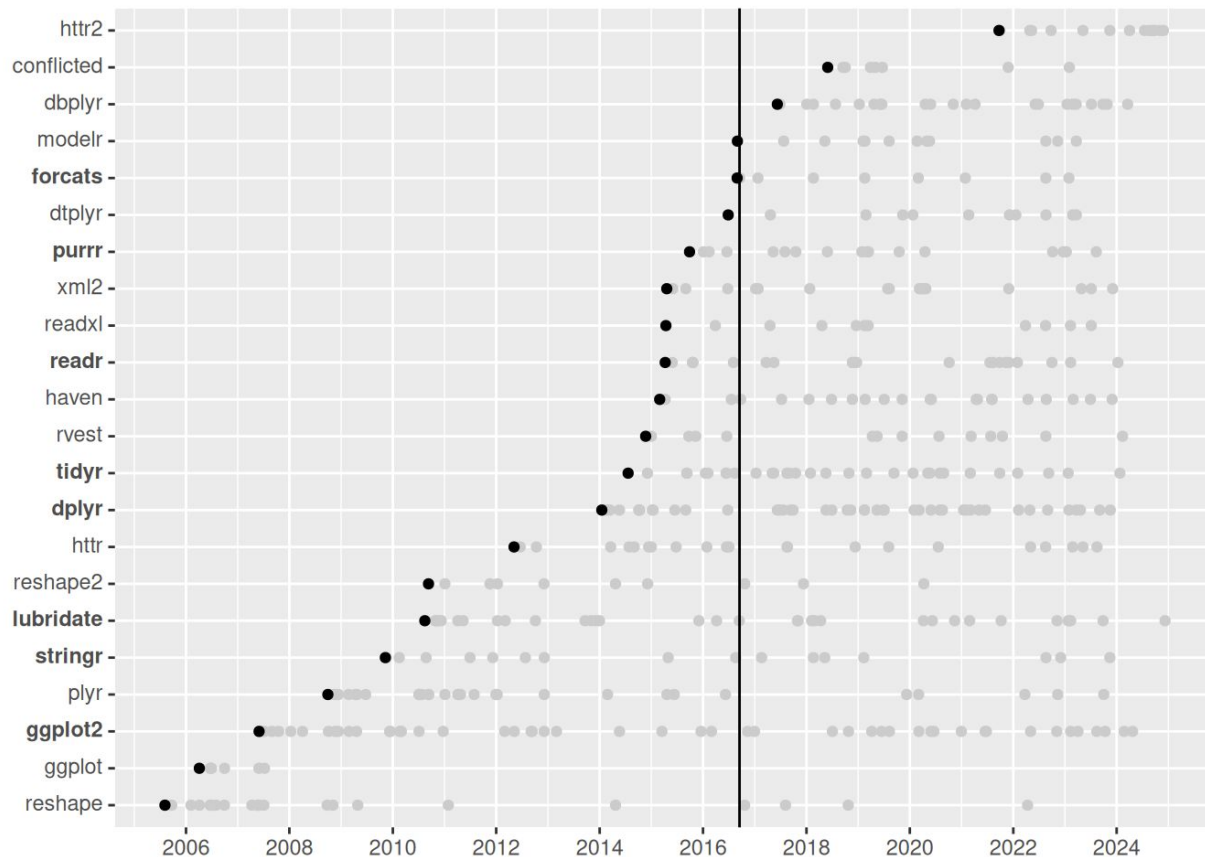
Packages designed for data science that make analysis more intuitive.

Super powerful for wrangling and data viz.



History of the tidyverse

"The tidyverse is just one way to get the job done and I don't think it's wrong to use other tools."



Import

readr
readxl
haven
xml2

Visualise

ggplot2



tidyverse.org

Tidy → Transform

tidyr

dplyr
forcats
hms

lubridate
stringr

Model

tidymodels



r4ds.had.co.nz

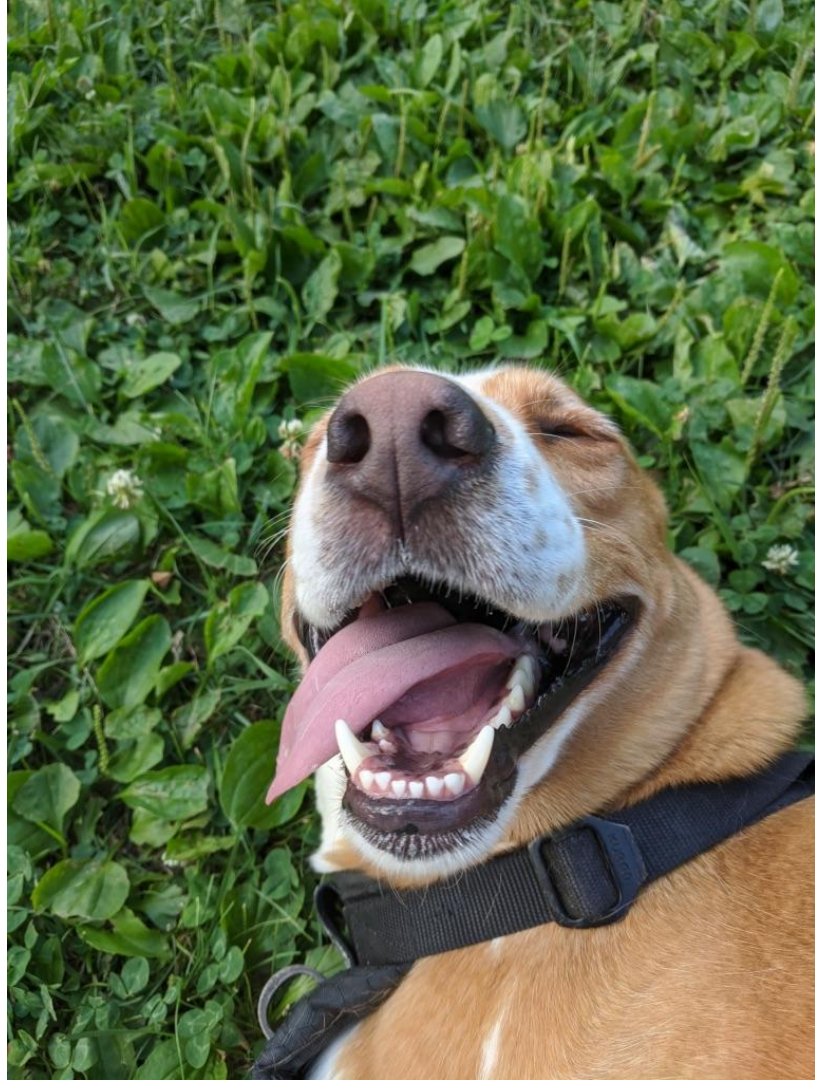
purrr
magrittr

Program

How many people feel
about data wrangling



How we feel about data
wrangling





About us



Carrie



Ava

Carrie Wright

Developing methods, tools, and educational resources to help researchers and others use data more rigorously, reproducibly, and ethically.



- Senior Staff Scientist at the Fred Hutchinson Cancer Research Center
- Affiliated faculty member at the Johns Hopkins Bloomberg School of Public Health (JHSPH)



Major Projects:

- Informatics Technology for Cancer Research (ITCR) Training Network (ITN)
- Open Case Studies

<https://carriewright11.github.io/>

BioOCS

Bloomberg
American
Health
Initiative

Free and
Open-
Source



Providing Real-World Data
Analysis Example Guides



Experiential

Evaluating real-world problems with realistic data in a hands-on experience



Complete

Demonstrating analyses from start to finish - emphasizing data preparation



Peer-reviewed

Our case studies have been reviewed by subject matter, data science, and statistics experts

"It is always so challenging to find datasets that allow students to explore rich and complex questions. The Open Case Studies project provides an **absolute gold mine of resources**".

<https://www.opencasestudies.org/>

**Ava Hoffman**

she/her

Senior Scientist at Fred Hutch Cancer Center studying genetics, ecology, and data science education.

📍 Baltimore, USA

🏢 Fred Hutch Cancer Center

✉️ Email

🔗 Google Scholar

🆔 ORCID

🐙 GitHub

🦋 Bluesky

<https://www.avahoffman.com/>

Hello!

Thanks for visiting! I'm currently a Senior Scientist at Fred Hutch Cancer Center studying genetics, ecology, and data science education.

I lead several education grant programs through National Institutes of Health, including [Data Science for Environmental Health](#), [Genomic Education Modules \(GEMs\)](#), and [Biomedical Open Case Studies](#). I'm also deeply involved with the [Genomic Data Science Community Network](#) and [the AnVIL](#), a cloud computing platform. At the core of my work is centering community while furthering data-intensive education and research.

In addition to my education related research, I study the evolutionary ecology and genomics of plants. Some of my work has focused on drought responses in prairie grasses and the evolution of weedy species in urban environments. Recently, I've been very involved in the [BioDIGS Research Consortium](#), a program that organizes under-resourced institutions to better understand soil metagenomic biodiversity and ecological patterns across the USA.

I'm in the process of revamping my website, so things look a bit messy right now. Please check later for a more polished version :)

Why this class



**LEARNED BASE
R AS A GRAD STUDENT**



**THEN DISCOVERED
THE TIDYVERSE**

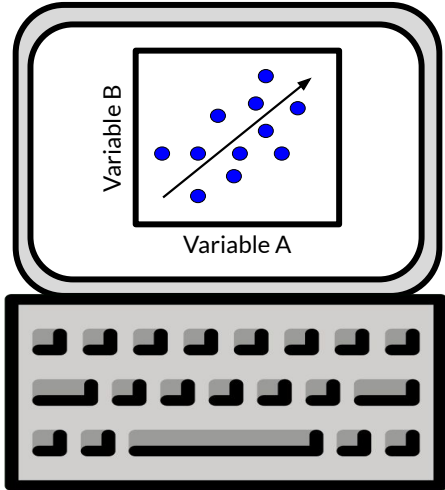
imgflip.com



But also...

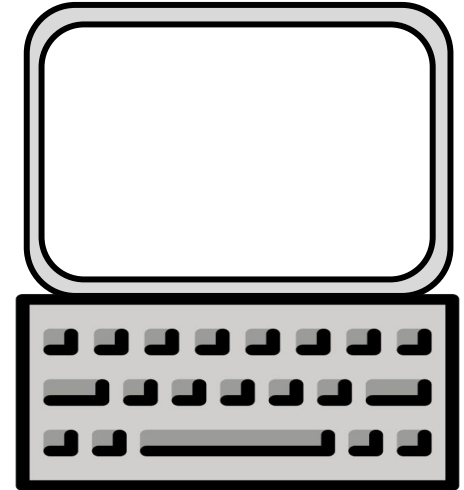
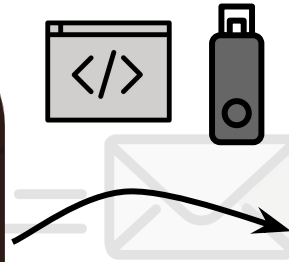
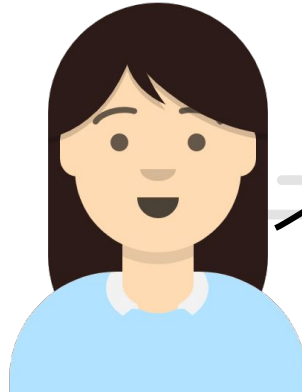
Data wrangling using more error-prone ways can lead to less transparent science

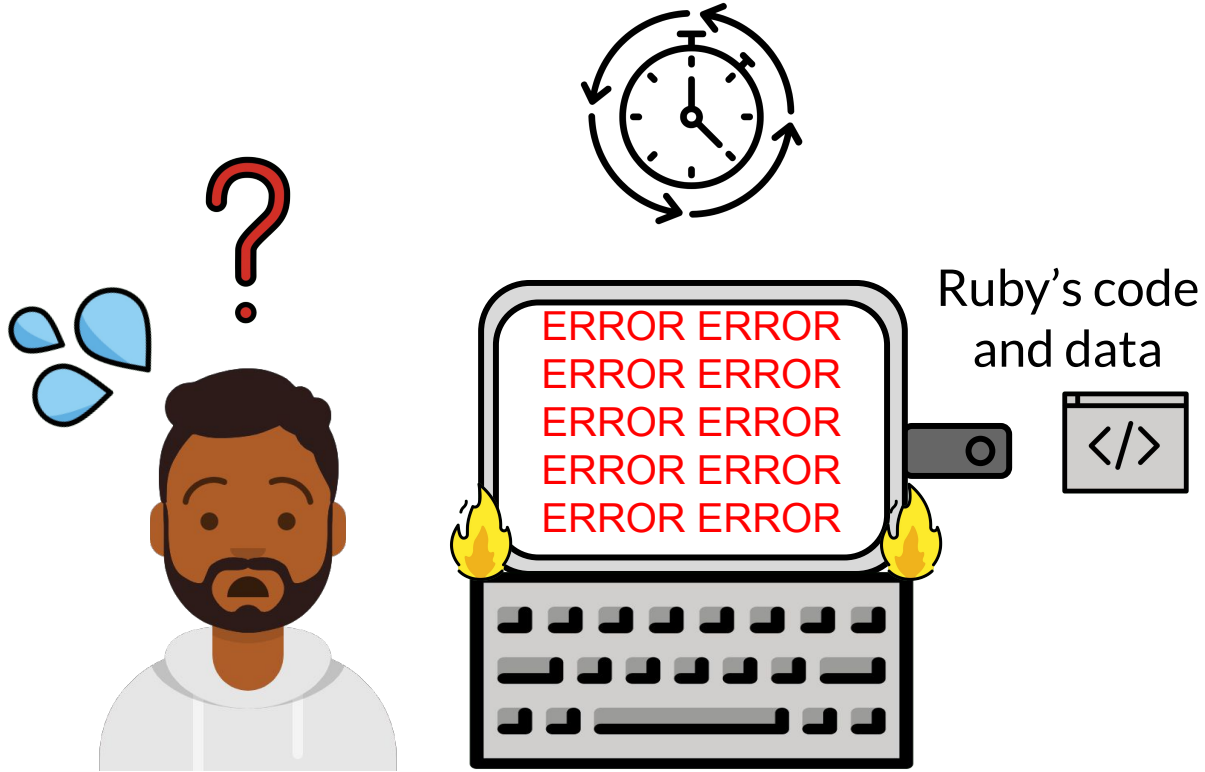
✓ $R^2 = 0.893$

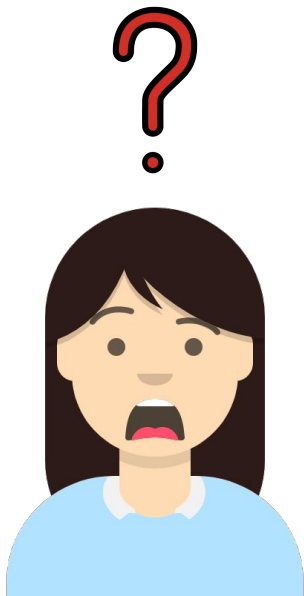


This code runs well on my computer, let me email it to you!

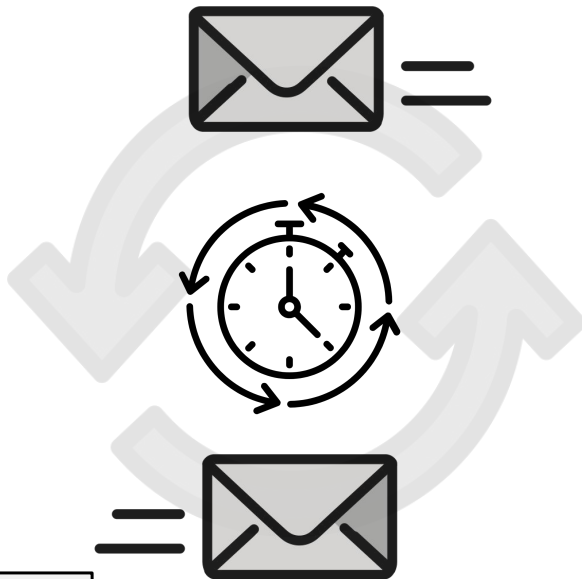
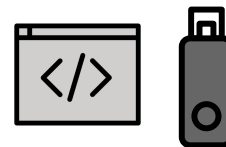
So exciting!



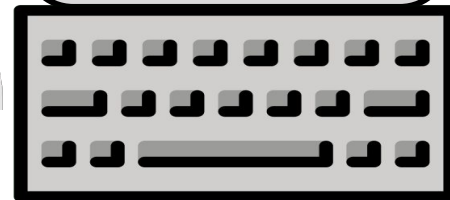




Re:Re:Re: Data
Hi Ruby, I don't understand
what this code is supposed
to be doing...

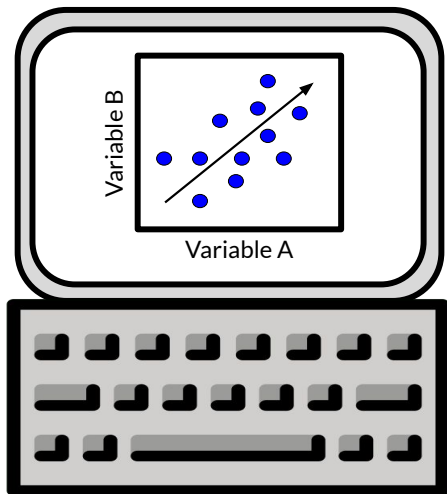


Error: file path
"Ruby's
computer/Ruby's
file/final_version10.
R" not found

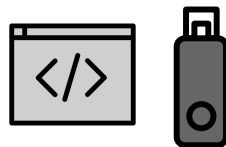


Re:Re:Re: Data
Hi Avi, It works for me?

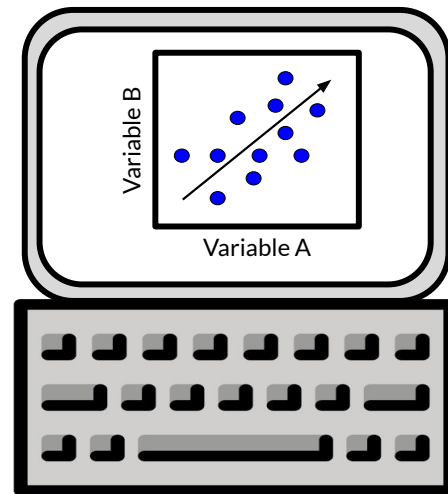
✓ $R = 0.893$

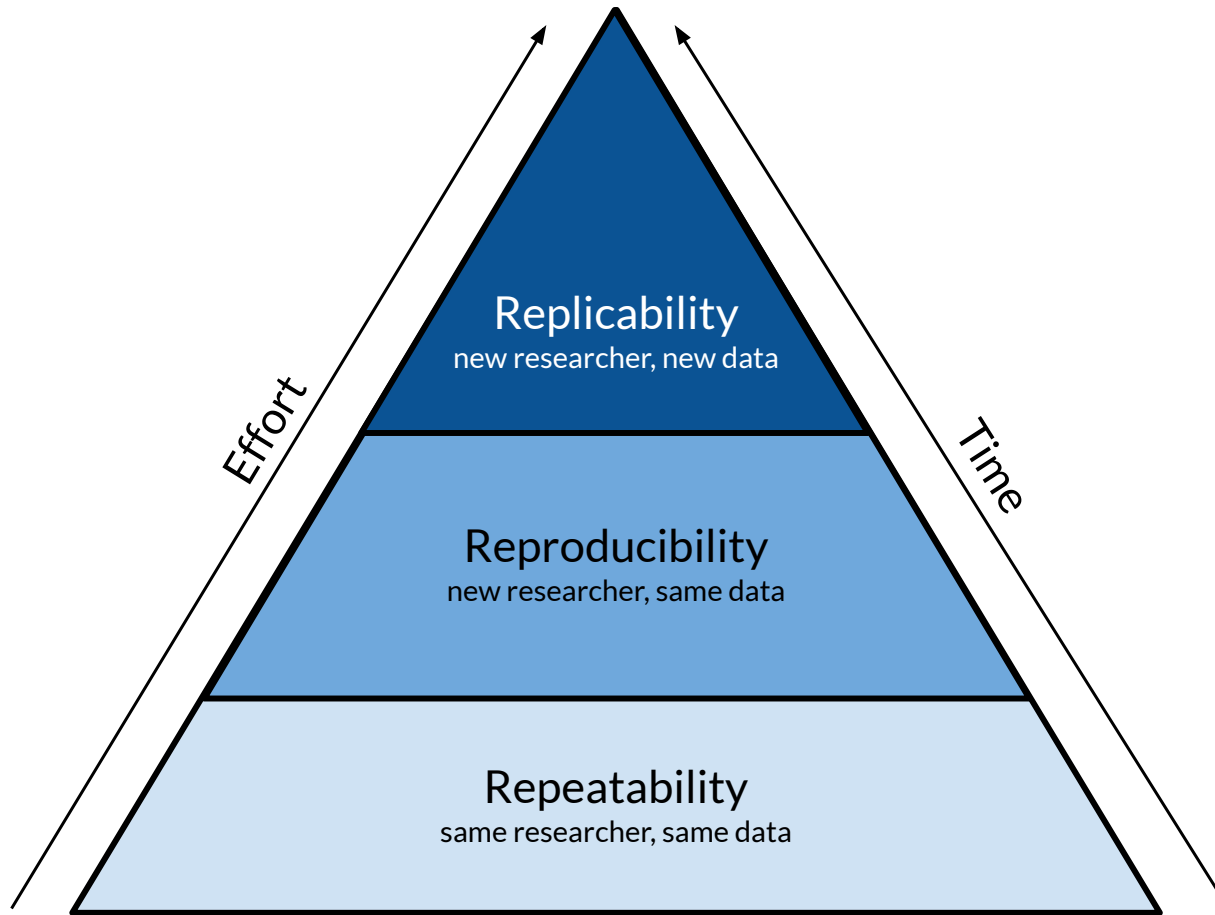


Ruby's code
and data

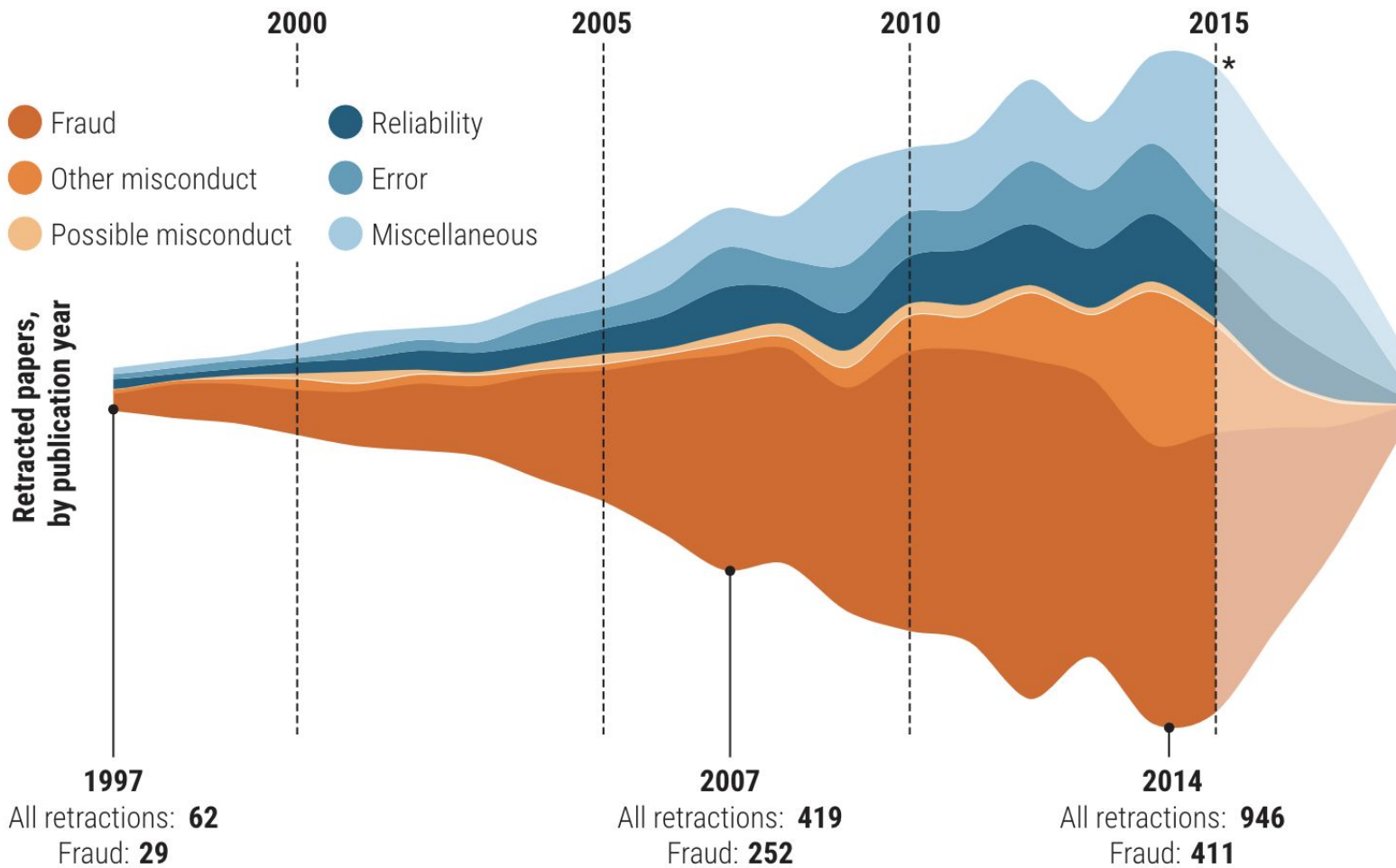


⊘ $R = 0.891$





Transparent Data Wrangling is Required
for Transparent Data Science



Dataset Error:

Data Errors are One of the Leading Causes of Retractions

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10485848/>

Dataset Error:

Data Errors are One of the Leading Causes of Retractions

In the case of Kufner et al.'s study on the smoking paradox in ischemic stroke patients, the authors acknowledged a significant error in their dataset labeling. This error led to a gross misrepresentation of the number of individuals who had received intra-arterial thrombolysis treatment, undermining the validity of the study's main conclusion. This example underscores the need for meticulous data representation and cleaning in research, highlighting the potential implications of errors and, in the occurrence of errors, the need for adequate conduct, even as it showcases a good example of authors owning their mishaps and shedding light on them ([Kufner et al., 2022](#)).

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0251888>

The smoking paradox in ischemic stroke patients treated with intra-arterial thrombolysis in combination with mechanical thrombectomy–VISTA-Endovascular

Anna Kufner , Huma Fatima Ali, Martin Ebinger, Jochen B. Fiebach, David S. Liebeskind, Matthias Endres, Bob Siegerink, on behalf of the VISTA-Endovascular Collaborators 

Published: May 20, 2021 • <https://doi.org/10.1371/journal.pone.0251888>

Article	Authors	Metrics	Comments	Media Coverage	Peer Review
⌵					

Retraction

Abstract

Introduction

Methods

Results

Discussion

Supporting information

Acknowledgments

References

Reader Comments

Figures

⚠ Retraction

After this article [1] was published, the authors became aware of a dataset error that renders the article's conclusions invalid.

Specifically, due to data labelling and missing information issues, the 'IAT' data reflect intra-arterial (IA) treatment rather than the more restricted treatment type of IA-thrombolysis. Further investigation of the dataset revealed that only 24 individuals in the study population received IA-thrombolysis, instead of N = 216 as was reported in [1]. Hence, the article's main conclusion is not valid or reliable as it is based on the wrong data.

Furthermore, due to the small size of the IA-thrombolysis-positive group, the dataset is not sufficiently powered to address the research question.

In light of the above concerns, the authors retract this article.

All authors agree with retraction.

12 Dec 2022: Kufner A, Ali HF, Ebinger M, Fiebach JB, Liebeskind DS, et al. (2022) Retraction: The smoking paradox in ischemic stroke patients treated with intra-arterial thrombolysis in combination with mechanical thrombectomy–VISTA-Endovascular. PLOS ONE 17(12): e0279276. <https://doi.org/10.1371/journal.pone.0279276> | [View retraction](#)

Retraction

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SCIENCEINSIDER | HEALTH

Two elite medical journals retract coronavirus papers over data integrity questions

Mysterious company Surgisphere declined to provide access to hospital data used to evaluate drugs in COVID-19 patients

4 JUN 2020 • BY [CHARLES PILLER](#), [KELLY SERVICK](#)

Older example - but of great consequence

Genomic signatures to guide the use of chemotherapeutics

Anil Potti^{1,2}, Holly K Dressman^{1,3}, Andrea Bild^{1,3}, Richard F Riedel^{1,2}, Gina Chan⁴, Robyn Sayer⁴, Janiel Cragun⁴, Hope Cottrill⁴, Michael J Kelley², Rebecca Petersen⁵, David Harpole⁵, Jeffrey Marks⁵, Andrew Berchuck^{1,6}, Geoffrey S Ginsburg^{1,2}, Phillip Febbo^{1,2,3}, Johnathan Lancaster⁴ & Joseph R Nevins^{1,2,3}

Using *in vitro* drug sensitivity data coupled with Affymetrix microarray data, we developed gene expression signatures that predict sensitivity to individual chemotherapeutic drugs. Each signature was validated with response data from an independent set of cell line studies. We further show that many of these signatures can accurately predict clinical response in individuals treated with these drugs. Notably, signatures developed to predict response to individual agents, when combined, could also predict response to multidrug regimens. Finally, we integrated the chemotherapy response signatures with signatures of oncogenic pathway deregulation to identify new therapeutic

ARTICLE LINKS

- ▶ Supplementary info

ARTICLE TOOLS

- ✉ Send to a friend
- 📄 Export citation
- 📄 Export references
- 🔒 Rights and permissions
- 📄 Order commercial reprints

SEARCH PUBMED FOR

- ▶ Anil Potti
- ▶ Holly K Dressman
- ▶ Andrea Bild
- ▶ Richard F Riedel

<https://doi.org/10.1038/nm1491>

Nature Medicine **12**, 1294–1300 (2006) | [Cite this article](#)

5843 Accesses | **426** Citations | **95** Altmetric | [Metrics](#)



i A [Retraction](#) to this article was published on 07 January 2011

i A [Corrigendum](#) to this article was published on 01 August 2008

i A [Corrigendum](#) to this article was published on 01 November 2007

Individual Researchers

Anil Potti's misbehavior is at the center of the case. Prior to ORI's conclusion of research misconduct, Joseph Nevins and Robert Califf had both said that it is highly likely that Potti intentionally fabricated or falsified data ([CBS News, 2012](#)). In addition, Baggerly, Coombes, and Wang had documented many instances of sloppy or careless data analysis, and Perez documented use of unreliable predictors and omission of data not showing desired results. The negative impact of such sloppy and careless practices on the ability to replicate results and ultimately on patient care might be similar to the impact of fabrication or falsification.

<https://www.ncbi.nlm.nih.gov/books/NBK475955/>

When is Reproducibility an Ethical Issue? Genomics, Personalized Medicine, and Human Error

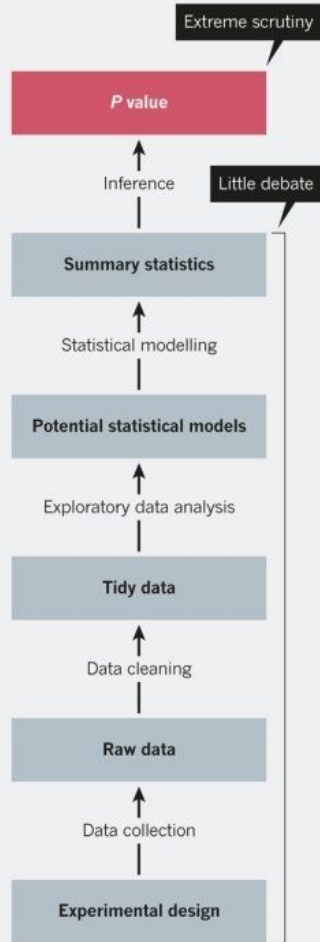
Keith Baggerly
Bioinformatics and Computational Biology
UT M. D. Anderson Cancer Center
kabagg@mdanderson.org

BIRS Workshop, Aug 14, 2013



DATA PIPELINE

The design and analysis of a successful study has many stages, all of which need policing.



- Most of the attention is on the last step
- This course is about all the steps that come before
- They are **critical** for getting things right

Reproducibility/Transparency is
Important



“We report the challenges confronted during a large-scale effort to replicate findings in cancer biology, and describe how improving **transparency and sharing** can make it easier to assess rigor and replicability and, therefore, to **increase research efficiency.**”

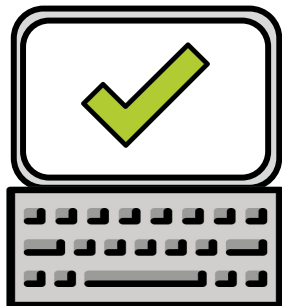
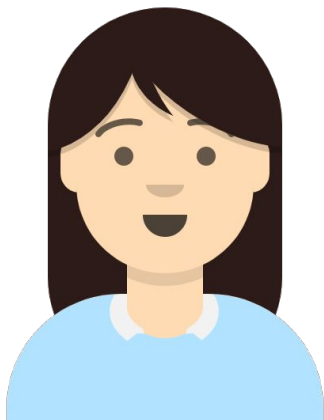
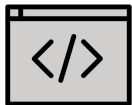
<https://elifesciences.org/articles/67995>

**Reproducibility saves
everyone time and effort!**



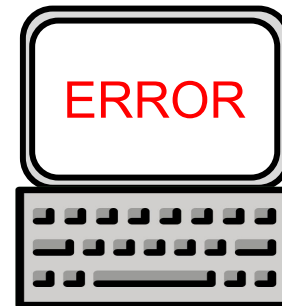
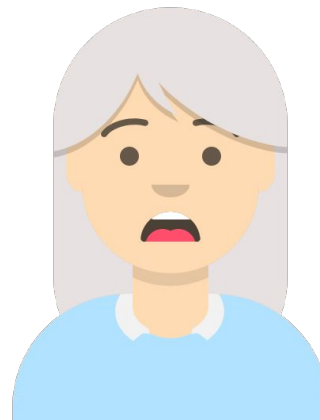
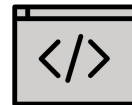
Now Ruby

Ruby's code



Future Ruby

Ruby's code



It saves your future self time and effort!

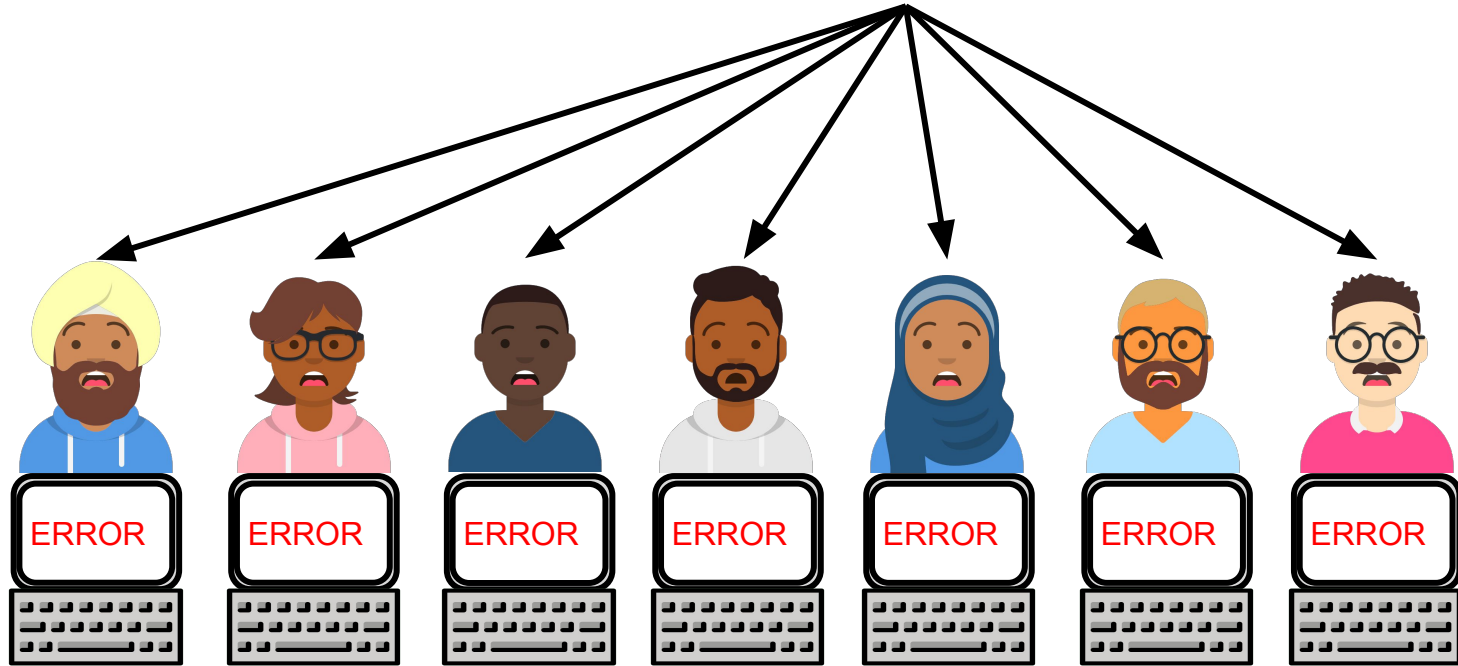
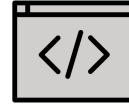
Your closest collaborator is
you six months ago, but you
don't reply to emails

- Karl Broman

http://kbroman.org/Tools4RR/assets/lectures/06_org_eda.pdf

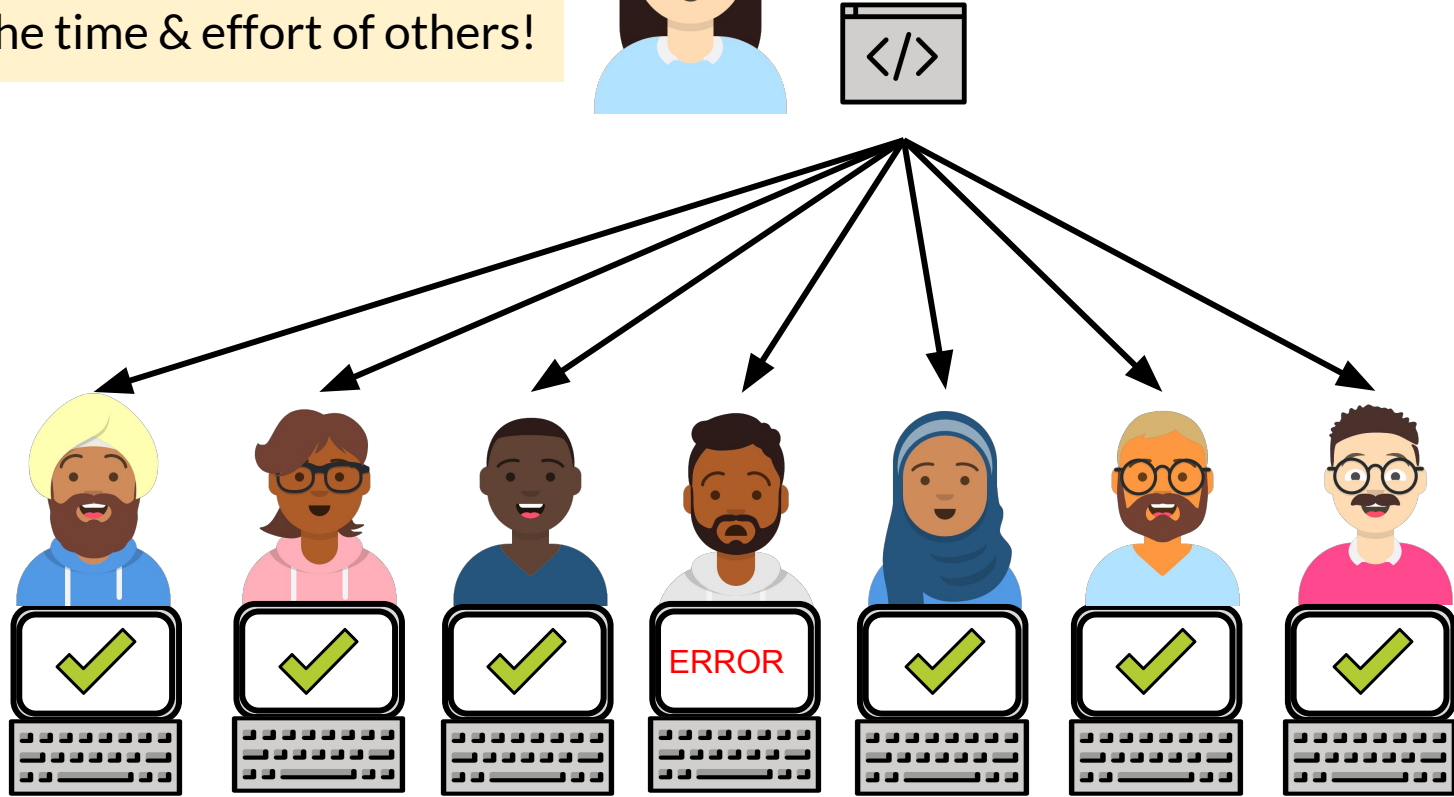


Ruby's code - **not as reproducible**

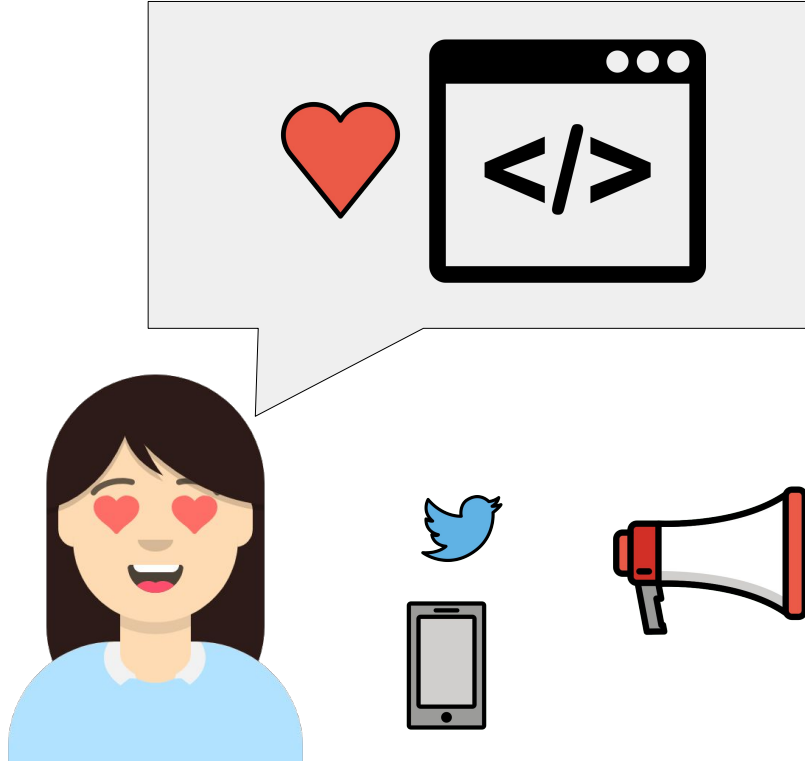


It saves the time & effort of others!

Ruby's code - **more reproducible**



A useable, well-documented analysis is more likely to be used and disseminated!



It improves trust and reuse!



⚠ Reproducibility $\neq$ Correctness ⚠

Reproducibility $\sim$ Consistency

You could be consistently
wrong in the same way....



However, being consistent and transparent is a necessary step for doing trustworthy science.



It makes it easier for you and others to determine if your work was correct.



Reproducible work is typically easier to update!



Reviewer 2:
Please redo your full analysis but
change this one detail.



What you wished data looked like

The screenshot shows a Microsoft Excel window with the file name "solutions-jun3.csv". The ribbon at the top includes tabs for New, Open, Save, Print, Import, Copy, Paste, Format, Undo, Redo, AutoSum, Sort A-Z, Sort Z-A, Gallery, Toolbox, Zoom, and Help. The font settings are Verdana, size 10. The spreadsheet displays a table with the following columns:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	id	problem_id	subject_id	start	stop	time_left	answer									
2	1	498	17	1307119989	1307120016	2369	A									
3	2	150	15	1307119991	1307120009	2376	D									
4	3	313	16	1307119994	1307120009	2376	E									
5	4	12	13	1307119995	1307120019	2366	B									
6	5	273	14	1307119996	1307120028	2357	A									
7	6	101	19	1307119996	1307120021	2364	B									
8	7	105	18	1307119998	1307120048	2337	B									
9	8	162	12	1307120004	1307120042	2343	C									
10	9	70	15	1307120011	1307120038	2347	C									
11	10	300	16	1307120012	1307120092	2293	B									
12	11	494	17	1307120017	1307120075	2310	D									
13	12	357	13	1307120021	1307120118	2267	A									
14	13	522	19	1307120025	1307120152	2233	D									
15	14	232	14	1307120030	1307120158	2227	C									
16	15	344	15	1307120041	1307120117	2268	B									
17	16	160	17	1307120079	1307120249	2136	D									
18	17	516	16	1307120094	1307120159	2226	B									
19	18	472	12	1307120119	1307120170	2215	A									
20	19	43	15	1307120122	1307120140	2245	C									
21	20	353	13	1307120144	1307120199	2186	C									
22	21	218	15	1307120152	1307120272	2113	E									
23	22	69	16	1307120163	1307120188	2197	D									
24	23	562	16	1307120190	1307120301	2084	D									
25	24	121	19	1307120253	1307120294	2091	E									
26	25	297	15	1307120277	1307120342	2043	B									
27	26	495	13	1307120281	1307120353	2032	E									
28	27	94	14	1307120288	1307120343	2042	E									
29	28	22	18	1307120310	1307120365	2020	C									
30	29	64	19	1307120310	1307120385	2000	B									
31	30	502	16	1307120323	1307120336	2049	B									
32	31	44	16	1307120339	1307120352	2033	A									
33	32	315	14	1307120348	1307120362	2023	B									
34	33	385	15	13												

What it actually looks like

<http://healthdesignchallenge.com/>

```
@HWI-EAS121:4:100:1783:550#0/1
CGTTACGAGATCGGAAGAGCGGTTTCAGCAGGAATGCCGAGACGGATCTCGTATGCCGTCTGCTGCGTGACAAGACAGGGG
+HWI-EAS121:4:100:1783:550#0/1
aaaaa`b_aa`aa`YaX]aZ`aZM^Z]YRa]YSG[[ZREQLHESDHNDHNMEEDDMPENITKFLFEEDDDHEJQMEDDD
@HWI-EAS121:4:100:1783:1611#0/1
GGGTGGGCATTTCCACTCGCAGTATGGGTTGCCGCACGACAGGCAGCGGTCAGCCTGCGCTTTGGCCTGGCCTTCGGAAA
+HWI-EAS121:4:100:1783:1611#0/1
a```\__`_````a``a``a_`_]a_]`]\`a_____`_^^`]X]_]XTV\_]]NX_XVX]]_TTTTG[VTHPN]VFDZ
@HWI-EAS121:4:100:1783:322#0/1
CGTTTATGTTTTTGAATATGTCTTATCTTAACGGTTATATTTTAGATGTTGGTCTTATTCTAACGGTCATATATTTTCTA
+HWI-EAS121:4:100:1783:322#0/1
abaa`^aaaaabbbbaababbbbbbb`bbbb`bbbbbbbbb`bbbaV^_a``a``]``aT]a__V\]]_]`a`]a_abbaV__
@HWI-EAS121:4:100:1783:1394#0/1
GGGTCTTTATTGGTCTGGTGATCCCCCATATTCTCCGGTTGTGTGGTTTAACCGATCATCGCGCATTACTTCCCGGCTGC
+HWI-EAS121:4:100:1783:1394#0/1
```[aa\b^^[]aabbb][`a_abbb`a``bbbbbabababaaaab_VZa_^__bab_X`[a\HV_[_][^_X\T_VQQ
@HWI-EAS121:4:100:1783:207#0/1
CCCTGGGAGATCGGAAGAGCGGTTTCAGCAGGAATGCCGAGACCGATCTCGTATGCCGTCTTCTGCTTGAAAAAAAAAACA
+HWI-EAS121:4:100:1783:207#0/1
abba`Xa\`^\`aa]ba__bba[a_O_a`aa`a`a]^V]X_a^YS\R_\H_[]\ZTDUZZUSOPX]]POP\GS\WSHHD
@HWI-EAS121:4:100:1783:455#0/1
GGGTAATTCAGGGACAATGTAATGGCTGCACAAAAAATACATCTTTCATGTTCCATTGCACCATTGACAAATACATATT
+HWI-EAS121:4:100:1783:455#0/1
abb_babbabababbbbbbbbbbbbbbba\`b`\abbbabbbbabbbbbbbaabbbbb`bb`ab_O_bab_Q_bbabaa_a
```



# What it actually looks like

----- ALLERGIES -----		----- MEDICATION HISTORY -----	
Last Updated: 01 Dec 2011 @ 0851		Last Updated: 11 Apr 2011 @ 1737	
Allergy Name:	TRIMETHOPRIM	Medication:	AMLODIPINE BESYLATE 10MG TAB
Location:	DAYT29	Instructions:	TAKE ONE TABLET BY MOUTH TAKE ON GRAPEFRUIT JUICE--
Date Entered:	09 Mar 2011	Status:	Active
Reaction:		Refills Remaining:	3
Allergy Type:	DRUG	Last Filled On:	20 Aug 2010
Drug Class:	ANTI-INFECTIVES,OTHER	Initially Ordered On:	13 Aug 2010
Observed/Historical:	HISTORICAL	Quantity:	45
Comments:	The reaction to this allergy was MILD (NO SQUELAE)	Days Supply:	90
		Pharmacy:	DAYTON
		Prescription Number:	2718953
Allergy Name:	TRAMADOL	Medication:	IBUPROFEN 600MG TAB
Location:	DAYT29	Instructions:	TAKE ONE TABLET BY MOUTH FOUR TI
Date Entered:	09 Mar 2011	Status:	Active
Reaction:	URINARY RETENTION	Refills Remaining:	3
Allergy Type:	DRUG	Last Filled On:	20 Aug 2010
Drug Class:	NON-OPIOID ANALGESICS	Initially Ordered On:	01 Jul 2010
Observed/Historical:	HISTORICAL	Quantity:	300
Comments:	gradually worsening difficulty emptying bladder		
ms tramadol pain medication becoming ab			





**Desiree Narango**

@DLNarango

Follow



Today's updates on #otherpeoplesdata:



8:56 AM - 22 Oct 2018

1 Like



And so we data wrangle

Raw & processed data

“Data are values of qualitative or quantitative variables, belonging to a set of items.”

“Data are values of qualitative or quantitative variables, belonging to a set of items.”

**Set of items:** Sometimes called the population; the set of objects you are interested in

“Data are values of qualitative or quantitative **variables**, belonging to a set of items.”

**Variables:** A measurement or characteristic of an item

“Data are values of **qualitative** or **quantitative** variables, belonging to a set of items.”

**Qualitative:** Country of origin, sex, treatment

**Quantitative:** Height, weight, blood pressure

Data sharing



1. The raw data.
2. A tidy data set
3. A code book describing each variable and its values in the tidy data set.
4. An explicit and exact recipe you used to go from 1 -> 2,3



# *Journal of Statistical Software*

MMMMMM YYYY, Volume VV, Issue II.

<http://www.jstatsoft.org/>

## Tidy Data

**Hadley Wickham**  
RStudio



<https://twitter.com/hadleywickham>

---

### Abstract

A huge amount of effort is spent cleaning data to get it ready for analysis, but there has been little research on how to make data cleaning as easy and effective as possible. This paper tackles a small, but important, component of data cleaning: data tidying. Tidy datasets are easy to manipulate, model and visualise, and have a specific structure: each variable is a column, each observation is a row, and each type of observational unit is a table. This framework makes it easy to tidy messy datasets because only a small set of tools are needed to deal with a wide range of un-tidy datasets. This structure also makes it easier to develop tidy tools for data analysis, tools that both input and output tidy datasets. The advantages of a consistent data structure and matching tools are demonstrated with a case study free from mundane data manipulation chores.





# Code book

anything doesn't make sense.

Files:

**1 Demographics:** tab 1 is schizophrenia patients, tab 2 is controls.

A. Cohort: M = Mannheim (Germany), C = Cologne (Germany), H= Hopkins. We had a few of our own patients so we included them too.

B. patient identification number

C. Age at time of CSF collection

D. Gender

E. BMI

F. Ethnicity (mostly Caucasian)

G. Diagnosis: DSM/ICD-10 diagnosis

H. Group: control, schizophrenia, or prodromal. I don't think we have enough power to run them as three groups so I combined prodromal and schizophrenia. I'm not sure if this was ok. Is it appropriate to do a ttest for SZ?

I. Medication: mostly untreated

J. Education more or less than 13 years

K. current smoking status: yes or no

Variable names

Variable descriptions

Variable units

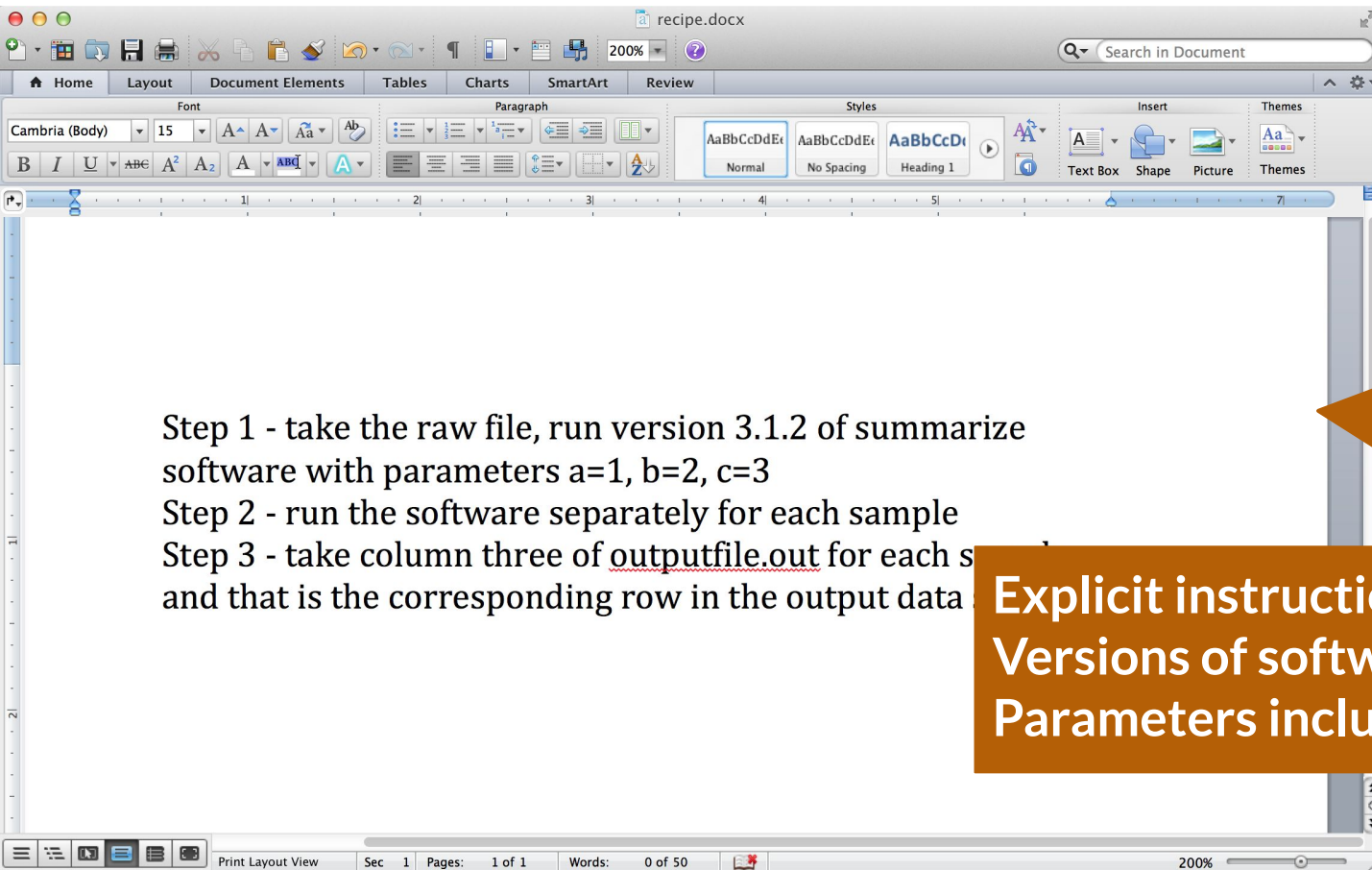
Study design quirks

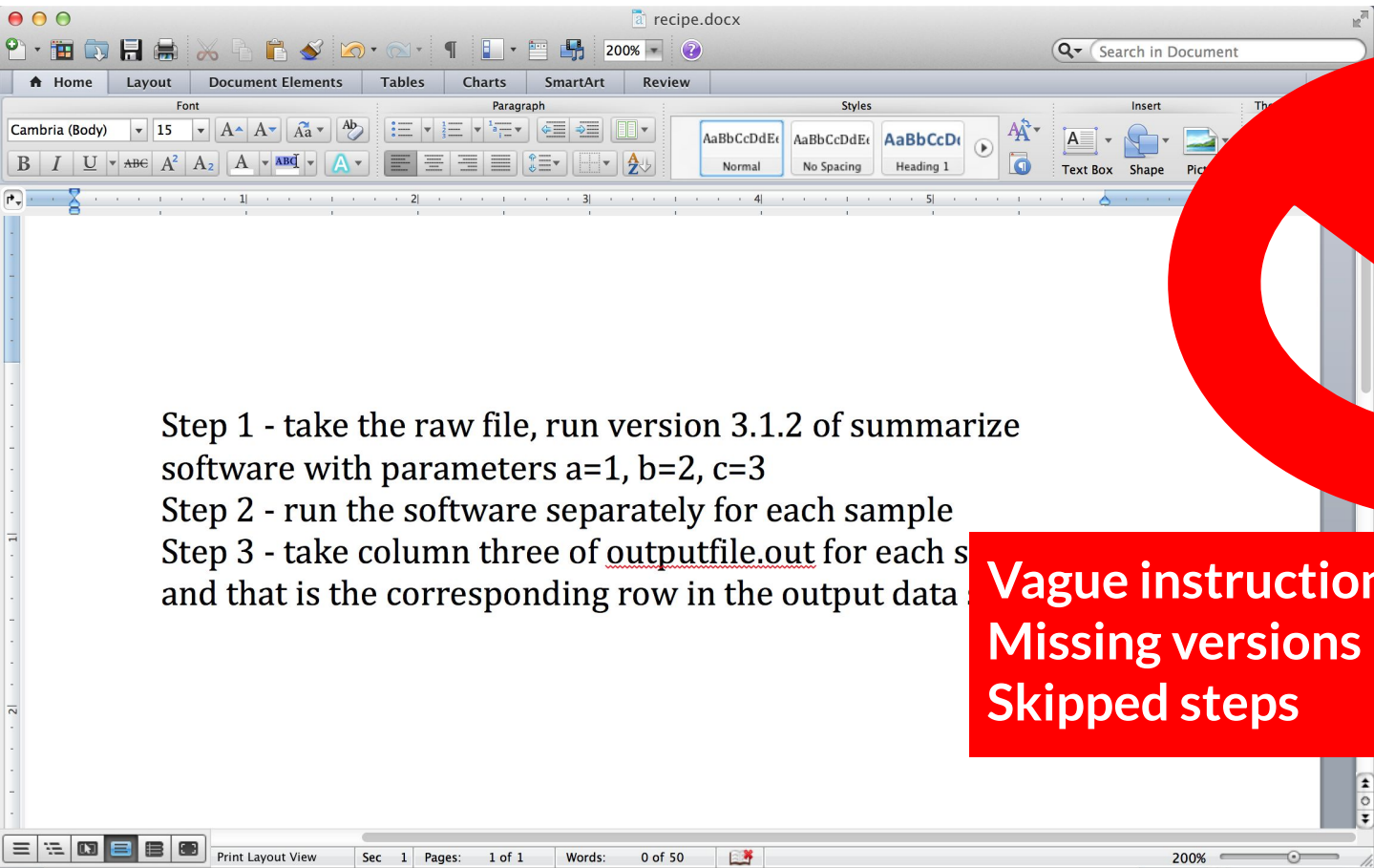
# Recipe

```
33 library(sva)
34 library(ffpe)
35 library(RColorBrewer)
36 library(corrplot)
37 library(limma)
38 trop = RSkittleBrewer('tropical')
39 ^
40
41
42 ▾ ## Load the data
43
44 You will need to download the GEUVADIS ballgown object from this site: https://github.com/ozgeezee/ballgown_code
45
46
47 ▾ ```{r loaddata,dependson="load"}
48 load("fpkm.rda")
49 pd = ballgown::pData(fpkm)
50 pd$dirname = as.character(pd$dirname)
51 ss = function(x, pattern, slot=1,...) sapply(strsplit(x, pattern), function(y) y[[slot]])
52 pd$IndividualID = ss(pd$dirname, "_", 1)
53 tfpkm = expr(fpkm)$trans
54 ^
55
56 ▾ ## Subset to non-duplicates
57
58 You will need the GEUVADIS quality control information and population information available from these
1:1 [f] (Top Level) ⇅
```

R/Python Code  
Input raw data -> output tidy  
No parameters







# Rules for Tidy Spreadsheets

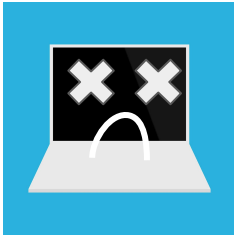
1. Be consistent
2. Choose good names for things
3. Write dates as YYYY-MM-DD
4. No empty cells
5. Put just one thing in a cell
6. Don't use font color or highlighting as data
7. Save the data as plain text files





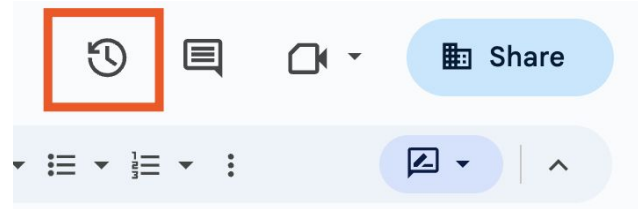
# Version Control!

Saving you from not knowing what or where your files are!



Saving you from losing your files when your computer dies or is stolen etc.!

Allowing you to track changes overtime and revisit them!



Like Google Doc history but better.

# Key principles of file naming for data science projects:

- Machine readable
- Human readable
- Be nicely ordered

When..	Be sure to...	So Do this...	Avoid this...	Why?
Naming variables (aka assigning column headers)	Use meaningful variable names	`AgeAtDiagnosis`	`ADx`	`ADx` is an unclear and uninformative abbreviation
Naming variables	Avoid spacing in column headers	`AgeAtDiagnosis`	`Age At Diagnosis`	Spacing in variable names makes the analyst's life more difficult
Naming variables	Use consistent capitalization	`AgeAtDiagnosis`	Using both `AgeAtDiagnosis` and `ageatdiagnosis`	Using consistent column names across tables/spreadsheets simplifies any merging the statistician may have to do.
Naming variables	Avoid using separators, but if it's necessary, use an underscore (`_`)	`IGF1` (or `IGF_1`)	`IGF.1`, `IGF-1`, `IGF/1`, `IGF,1`	Separators (commas, periods, hyphens, slashes, spaces etc.) often have different meanings in coding languages than they do in text. Avoiding them avoids error.
Coding variables	Avoid unnecessary spaces	`male`	`male `	That extra space after `male ` makes it different from `male` without a space.
Coding variables	Be consistent!	`male`	`Male`,`male`, and `M`	In the eyes of the statistician, `Male`,`male`, and `M` could be incorrectly perceived as three different values.
Coding variables	Be careful of spelling errors	`male`	`maale`	That extra `a` makes these two different categories.
Coding date and time	Use ISO 8601 coding	`YYYY-MM-DD`	`MM/DD/YY` and `Month Day, Year`	Consistency simplifies the analyst's life, and YYYY-MM-DD will not be misconstrued if opened in Excel.
Coding missing data	Not leave any cells blank and use a consistent value	`NA`	`0`, `-9`, red-highlighted blank cells, `.`; `.`; ...	Each cell should be filled with a consistent value. Pick a way to denote missingness (ideally `NA`) and stick with it. Avoid using numbers or punctuation to denote missing data.
Entering data	Stick to text and numbers	Convey all information with direct text/numerical entry	Using cell highlighting or font color to convey information	Your analyst may not use the same platform for analysis as you used for data entry, so avoiding font color and cell highlighting will minimize issues.
Generating an Excel file	Save the data in an appropriate format	Use one worksheet per table and save as CSV or text files	Multiple worksheets	Statisticians require this format to import your data onto other platforms.
Entering Data	Avoid entering unnecessary lines of text at the start	Start your first row with variable names	Adding lines of text	This violates the rules of tidy data and makes processing more difficult. Include this information in the "Code book" instead.
Opening files in Excel	Know and avoid its pitfalls	Consistently include one value per cell and be careful of date and time data.	Using macros, splitting cells, and merging cells	These formats are not amenable to data analysis on other platforms.

Bad Naming	Good Naming
2013 my report.md	2013_my_report.md
malik's_report.md	maliks_report.md
01_zoë_report.md	01_zoe_report.md
AdamHooverReport.md	adam-hoover-report.md
executivereportpepsiv1.md	executive_report_pepsi_v1.md

2018\_jan\_sales\_cust001\_prod001.md  
2017\_mar\_sales\_cust001\_prod001.md  
2016\_may\_sales\_cust001\_prod008.md  
2017\_jan\_sales\_cust120\_prod007.md  
2015\_oct\_sales\_cust034\_prod001.md  
2015\_oct\_sales\_cust034\_prod002.md

Year	Month	Type	Customer ID	Product ID
2018	jan	sales	001	001
2017	mar	sales	001	001
2016	may	sales	001	008
2017	jan	sales	120	007
2015	oct	sales	034	001
2015	oct	sales	034	002

Which one is better?

analysis.R

or

2017-exploratory\_analysis\_crime.R?

Which one is better?

05-21-2017-analysis-cust001.R

or

2017-05-21-analysis-cust001.R?



# Structure of a filename

processed\_pvalue\_data\_from\_pubmed\_oct24.rda

What did I do to this data

`processed_pvalue_data_from_pubmed_oct24.rda`

What kind of data is this?

processed\_**pvalue\_data**\_from\_pubmed\_oct24.rda

Where did it come from?

processed\_pvalue\_data\_from\_pubmed\_oct24.rda

When did I get it?

processed\_pvalue\_data\_from\_pubmed\_oct24.rda

Underscores/slashes not dots/whitespace

processed\_pvalue\_data\_from\_pubmed\_oct24.rda

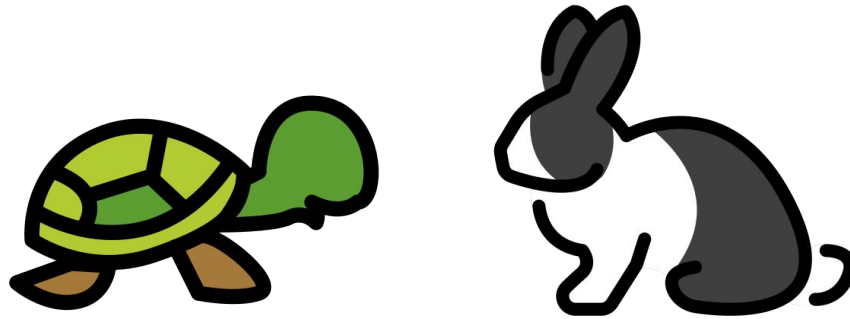
# Consistency is the main rule

processed\_pvalue\_data\_from\_pubmed\_oct24.rda  
raw\_pvalue\_data\_from\_pubmed\_oct24.rda

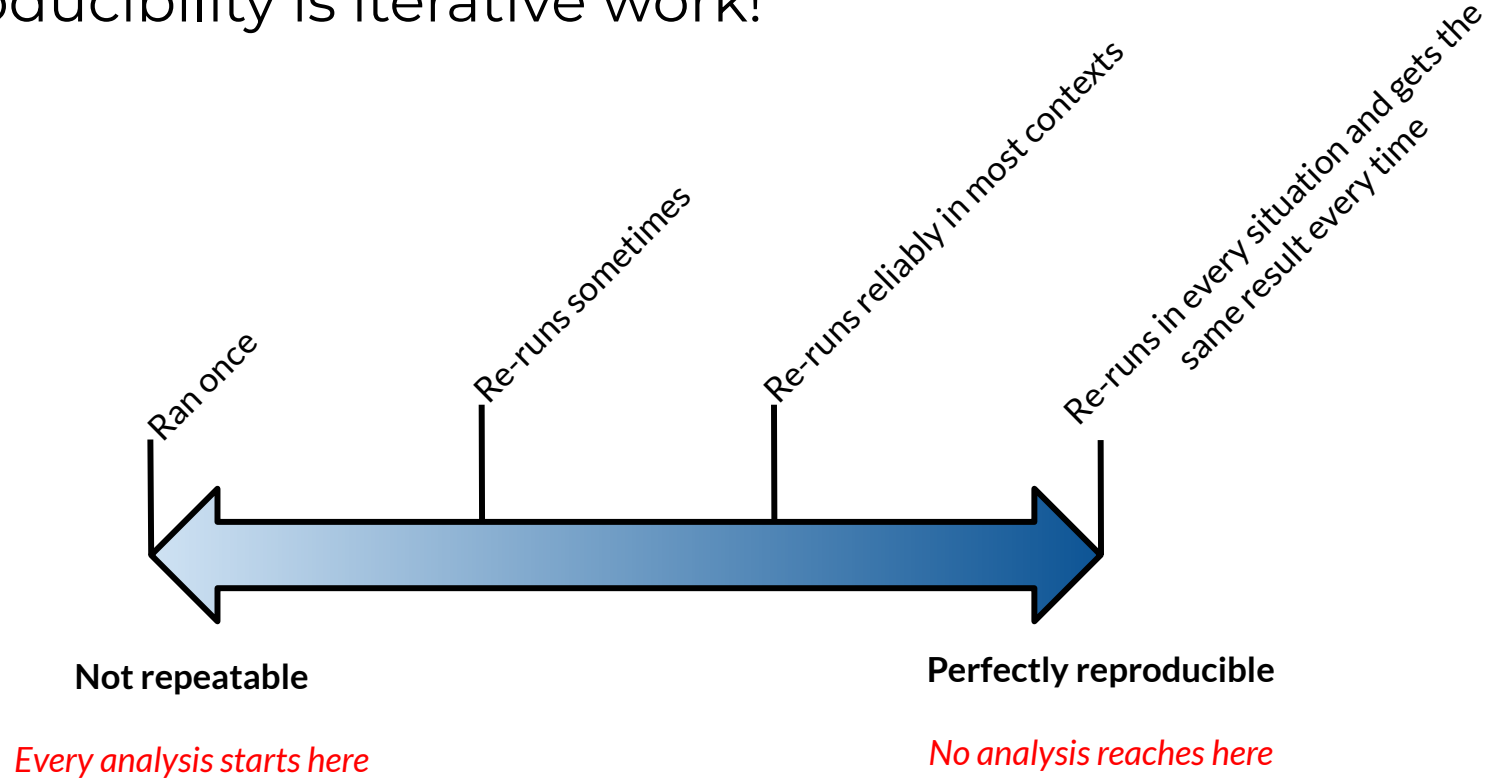
Organize thyself



Reproducibility is a tortoise's game - it's an incremental and slow process *but* **it has high payoffs!**



# Reproducibility is iterative work!



# Documentation that every project should have!

## 1. READMEs

AI can help, but check everything!

- a. Background knowledge
- b. Usage info
- c. Software requirements to run the thing
- d. Basics on how the files are organized

## 2. Code annotations:

- a. Explain historical decisions
- b. Explain “quirks” of the code
- c. Say where more development is needed (TODO)
- d. Summarize the goals!



# This is the README file for my\_first\_project

---

Last updated: 02-Mar-2018

The folders in this project are:

- *data* - is the folder where you can find all the collected data.
- *figures* - is where you can find all the plots, data pictures, and other images.
- *code* - is where you can find code files for collecting, cleaning up, or analyzing data.
- *products* - is where you can find reports, presentations, or products

Data on crime is obtained from International Crime Data collected between 2015-2018 and is publicly available. Data on happiness is collected from the Survey of International Happiness.

Contributors:

- Jane Everyday Doe, [jane.everyday.doe@gmail.com](mailto:jane.everyday.doe@gmail.com)
- John Everyday Doe, [john.everyday.doe@gmail.com](mailto:john.everyday.doe@gmail.com)

Cite: Doe, J, and Doe, J, Sample Analysis Using Sample Data, Working Paper, 2018

"File organization and naming are powerful weapons against chaos."  
- Jenny Bryan





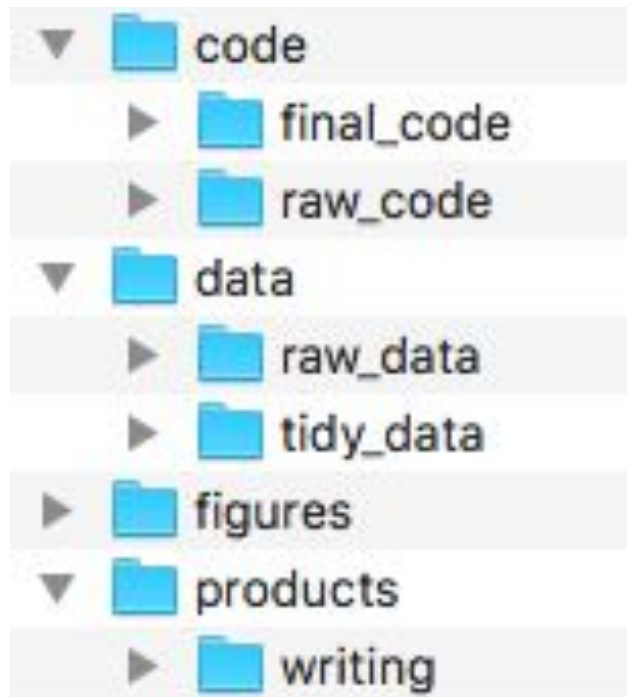


Name	
.DS_Store	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_A01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_A02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_A03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_B01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_B02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_B03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_C01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_C02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_C03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_D01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_D02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_D03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_E01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_E02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_E03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_F01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_F02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_F03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_G01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_G02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_G03.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_H01.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_H02.csv	
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_H03.csv	

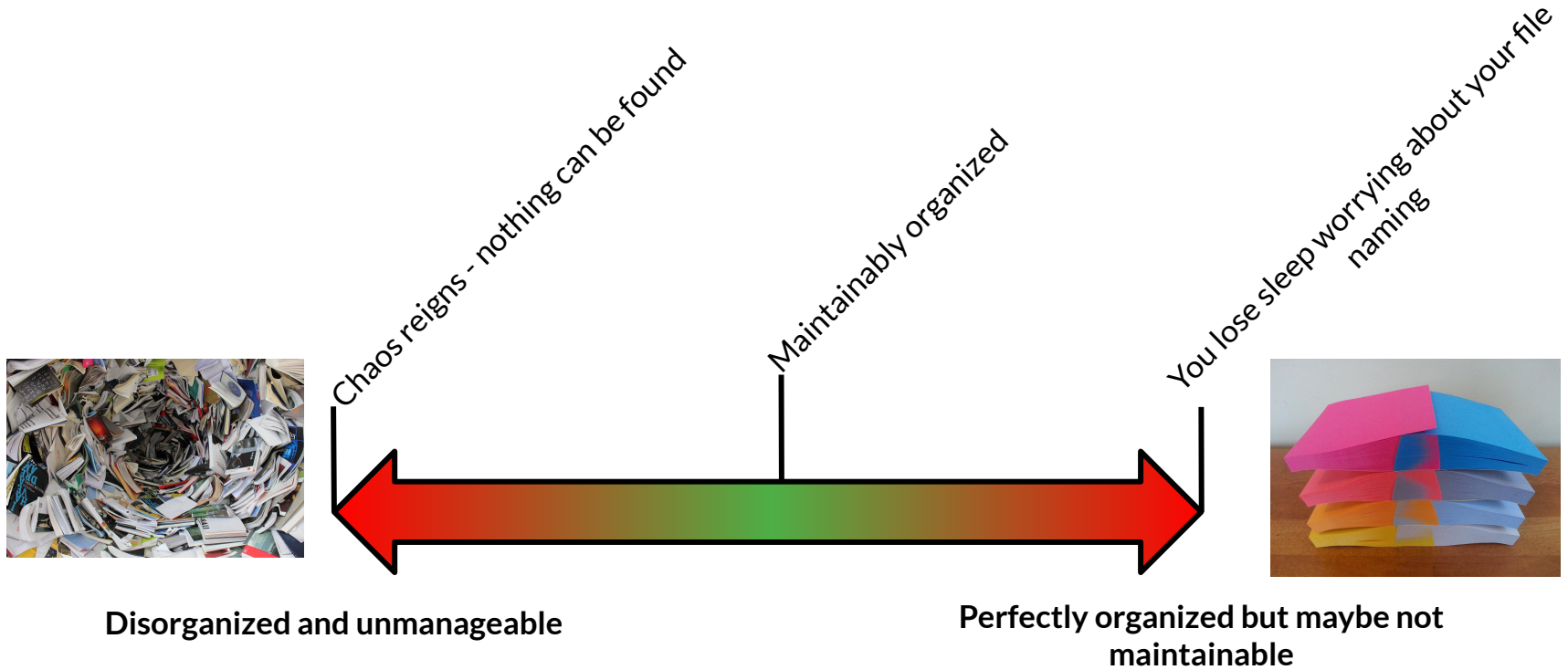
Slide via Jenny Bryan:

<http://www.slideshare.net/jenniferbryan5811/cm002-deep-thoughts>

2014-02-26\_BRAFWTNEGASSAY\_FFPEDNA-CRC-1-41\_A03.csv



# Your organizational system should work for you not the other way around!



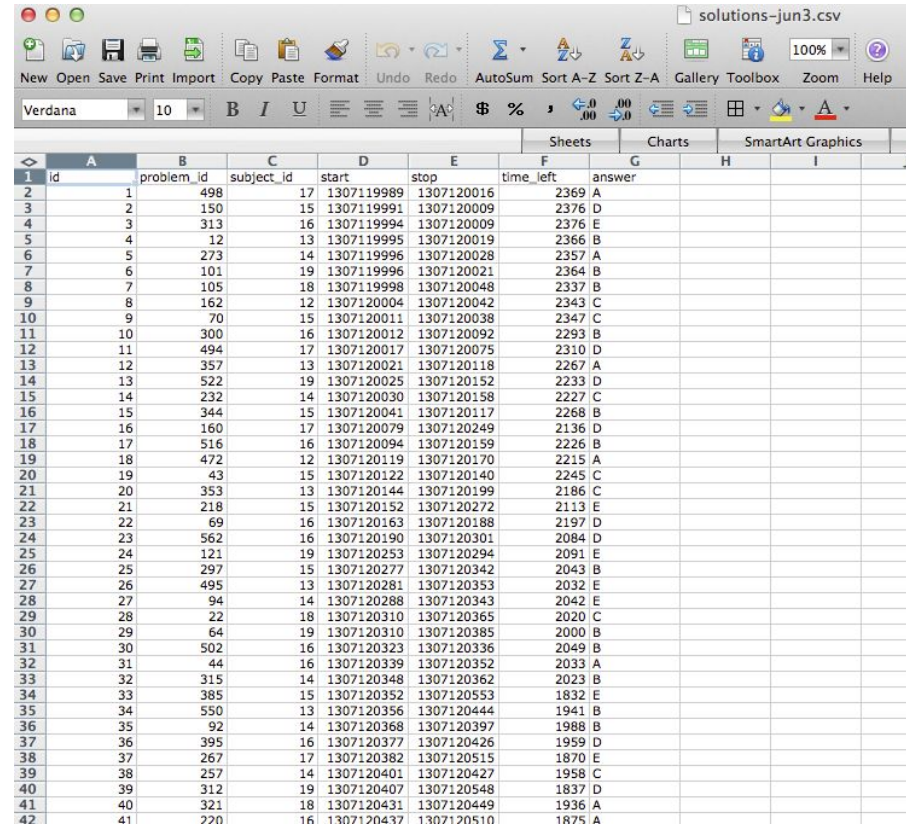


# Raw data

ALLERGIES		MEDICATION HISTORY	
Last Updated: 01 Dec 2011 @ 0851		Last Updated: 11 Apr 2011 @ 1737	
Allergy Name:	TRIMETHOPRIM	Medication:	AMLODIPINE BESYLATE 10MG TAB
Location:	DAYT29	Instructions:	TAKE ONE TABLET BY MOUTH TAKE ONE-HALF TABLET FOR GRAPEFRUIT JUICE--
Date Entered:	09 Mar 2011	Status:	Active
Reaction:		Refills Remaining:	3
Allergy Type:	DRUG	Last Filled On:	20 Aug 2010
Drug Class:	ANTI-INFECTIVES,OTHER	Initially Ordered On:	13 Aug 2010
Observed/Historical:	HISTORICAL	Quantity:	45
Comments:	The reaction to this allergy was MILD (NO SQUELAE)	Days Supply:	90
		Pharmacy:	DAYTON
		Prescription Number:	2718953
Allergy Name:	TRAMADOL	Medication:	IBUPROFEN 600MG TAB
Location:	DAYT29	Instructions:	TAKE ONE TABLET BY MOUTH FOUR TIMES A DAY WITH FOOD
Date Entered:	09 Mar 2011	Status:	Active
Reaction:	URINARY RETENTION	Refills Remaining:	3
Allergy Type:	DRUG	Last Filled On:	20 Aug 2010
Drug Class:	NON-OPIOD ANALGESICS	Initially Ordered On:	01 Jul 2010
Observed/Historical:	HISTORICAL	Quantity:	300
Comments:	gradually worsening difficulty emptying bladder		

# Processed data

- Processed data should be named so it is easy to see which script generated the data.
- The processing script - processed data mapping should occur in the README
- Processed data should be tidy



	A	B	C	D	E	F	G	H	I	J
	id	problem_id	subject_id	start	stop	time_left	answer			
1	1	498	17	1307119989	1307120016	2369	A			
2	2	150	15	1307119991	1307120009	2376	D			
3	3	313	16	1307119994	1307120009	2376	E			
4	4	12	13	1307119995	1307120019	2366	B			
5	5	273	14	1307119996	1307120028	2357	A			
6	6	101	19	1307119996	1307120021	2364	B			
7	7	105	18	1307119998	1307120048	2337	B			
8	8	162	12	1307120004	1307120042	2343	C			
9	9	70	15	1307120011	1307120038	2347	C			
10	10	300	16	1307120012	1307120092	2293	B			
11	11	494	17	1307120017	1307120075	2310	D			
12	12	357	13	1307120021	1307120118	2267	A			
13	13	522	19	1307120025	1307120152	2233	D			
14	14	232	14	1307120030	1307120158	2227	C			
15	15	344	15	1307120041	1307120117	2268	B			
16	16	160	17	1307120079	1307120249	2136	D			
17	17	516	16	1307120094	1307120159	2226	B			
18	18	472	12	1307120119	1307120170	2215	A			
19	19	43	15	1307120122	1307120140	2245	C			
20	20	353	13	1307120144	1307120199	2186	C			
21	21	218	15	1307120152	1307120272	2113	E			
22	22	69	16	1307120163	1307120188	2197	D			
23	23	562	16	1307120190	1307120301	2084	D			
24	24	121	19	1307120253	1307120294	2091	E			
25	25	297	15	1307120277	1307120342	2043	B			
26	26	495	13	1307120281	1307120353	2032	E			
27	27	94	14	1307120288	1307120343	2042	E			
28	28	22	18	1307120310	1307120365	2020	C			
29	29	64	19	1307120310	1307120385	2000	B			
30	30	502	16	1307120323	1307120336	2049	B			
31	31	44	16	1307120339	1307120352	2033	A			
32	32	315	14	1307120348	1307120362	2023	B			
33	33	385	15	1307120352	1307120553	1832	E			
34	34	550	13	1307120356	1307120444	1941	B			
35	35	92	14	1307120368	1307120397	1988	B			
36	36	395	16	1307120377	1307120426	1959	D			
37	37	267	17	1307120382	1307120515	1870	E			
38	38	257	14	1307120401	1307120427	1958	C			
39	39	312	19	1307120407	1307120548	1837	D			
40	40	321	18	1307120431	1307120449	1936	A			
41	41	220	16	1307120437	1307120510	1875	A			

- Step 1: slow down and document.
- Step 2: have sympathy for your future self.
- Step 3: have a system.

- Karl Broman

[http://kbroman.org/Tools4RR/assets/lectures/06\\_org\\_eda.pdf](http://kbroman.org/Tools4RR/assets/lectures/06_org_eda.pdf)

R + RStudio



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# The R Project for Statistical Computing

## Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

## News

- [The R Journal Volume 7/1](#) is available.
- [R version 3.2.1 \(World-Famous Astronaut\)](#) has been released on 2015-06-18.
- [R version 3.1.3 \(Smooth Sidewalk\)](#) has been released on 2015-03-09.
- [useR! 2015](#), will take place at the University of Aalborg, Denmark, June 30 - July 3, 2015.
- [useR! 2014](#), took place at the University of California, Los Angeles, USA June 30 - July 3, 2014.

RSTUDIO IDE

# The most trusted IDE for open source data science

RStudio is an integrated development environment (IDE) for R and Python. It includes a console, syntax-highlighting editor that supports direct code execution, and tools for plotting, history, debugging, and workspace management. RStudio is available in open source and commercial editions and runs on the desktop (Windows, Mac, and Linux).



SISBID\_Data-Wrangling - cleaning25 - RStudio

Go to file/function Addins

SISBID\_Data-Wrangling

Console Terminal Render Background Jobs

R 4.5.0 · ~/Documents/GitHub/Teaching/SISBID\_Data-Wrangling/

Environment History Connections Build Tutorial

Global Environment

Data

Variable	Obs.	Vars.
bike	1631	2
p	196	5
South_West	5763	12
ufo	88875	11
ufo_clean	88679	12

Values

Variable	Values
A	num [1:4] 1 2 3 NA
B	num [1:4] 1 2 3 4
test	num [1:4] 0 NA -1 NaN
url	"https://sisbid.github.io/Data-Wrangling..."

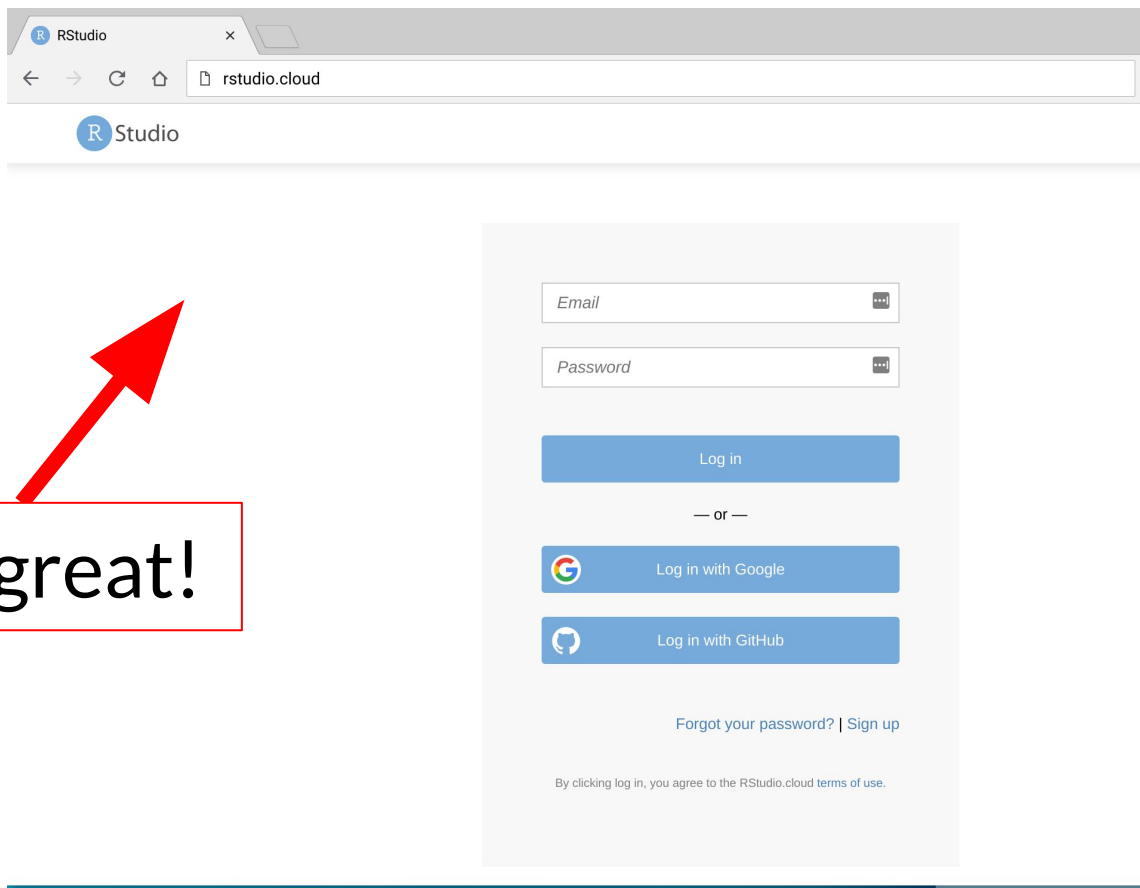
Files Plots Packages Help Git Viewer Presentation

Documents > GitHub > Teaching > SISBID\_Data-Wrangling > labs

Name	Size	Modified
..		
1000genomes.xlsx	967.1 KB	Aug 6, 2024, 1
20130606_sample_info.xlsx	967.2 KB	Aug 6, 2024, 1
advanced-io-lab-key.html	619.7 KB	Jul 15, 2025, 4
advanced-io-lab-key.R	2.5 KB	Aug 11, 2024,
advanced-io-lab-key.Rmd	2.9 KB	Jul 15, 2025, 4
advanced-io-lab.html	648.2 KB	Aug 6, 2024, 1
advanced-io-lab.R	1 KB	Aug 6, 2024, 1
advanced-io-lab.Rmd	2.1 KB	Aug 6, 2024, 1
Bike_Lanes_Long.csv	26.7 KB	Aug 6, 2024, 1
Bike_Lanes_Wide.csv	7 KB	Aug 6, 2024, 1
Bike_Lanes.csv	152.6 KB	Aug 6, 2024, 1
Circulator.csv	86.6 KB	Aug 6, 2024, 1
Circulator.rds	155.7 KB	Aug 6, 2024, 1
crashes_corrections.csv	268 B	Aug 6, 2024, 1

<https://www.rstudio.com/>

Also great!



RStudio

rstudio.cloud

R Studio

Email

Password

Log in

— or —

Log in with Google

Log in with GitHub

[Forgot your password?](#) | [Sign up](#)

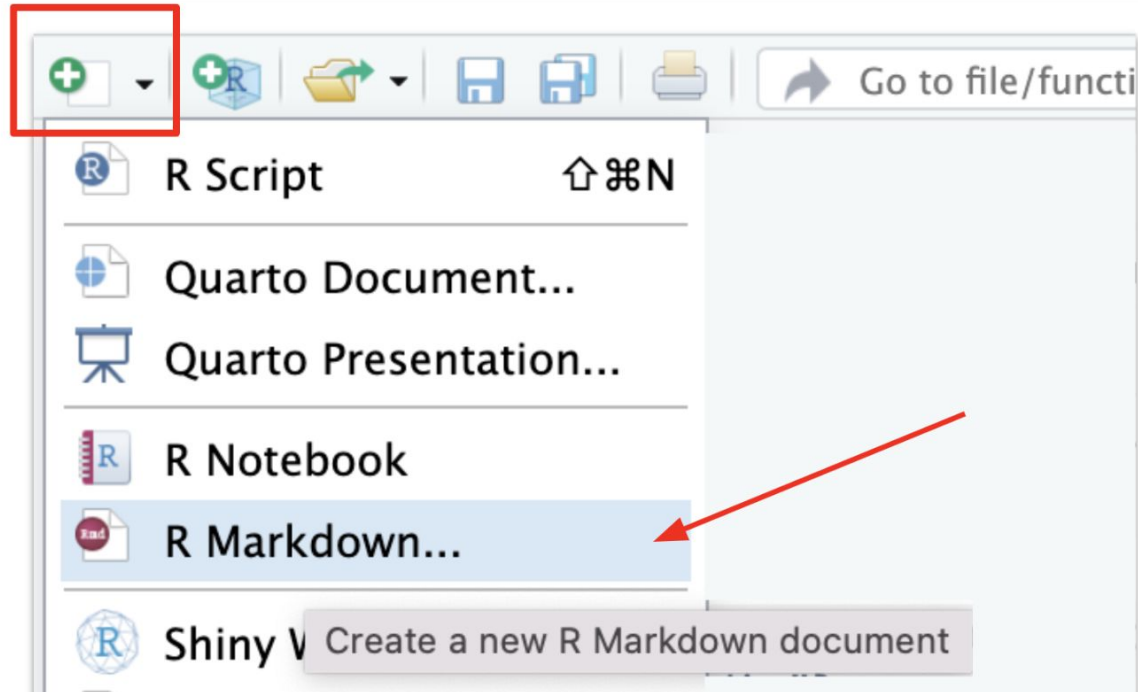
By clicking log in, you agree to the [RStudio.cloud terms of use](#).

<https://rstudio.cloud>



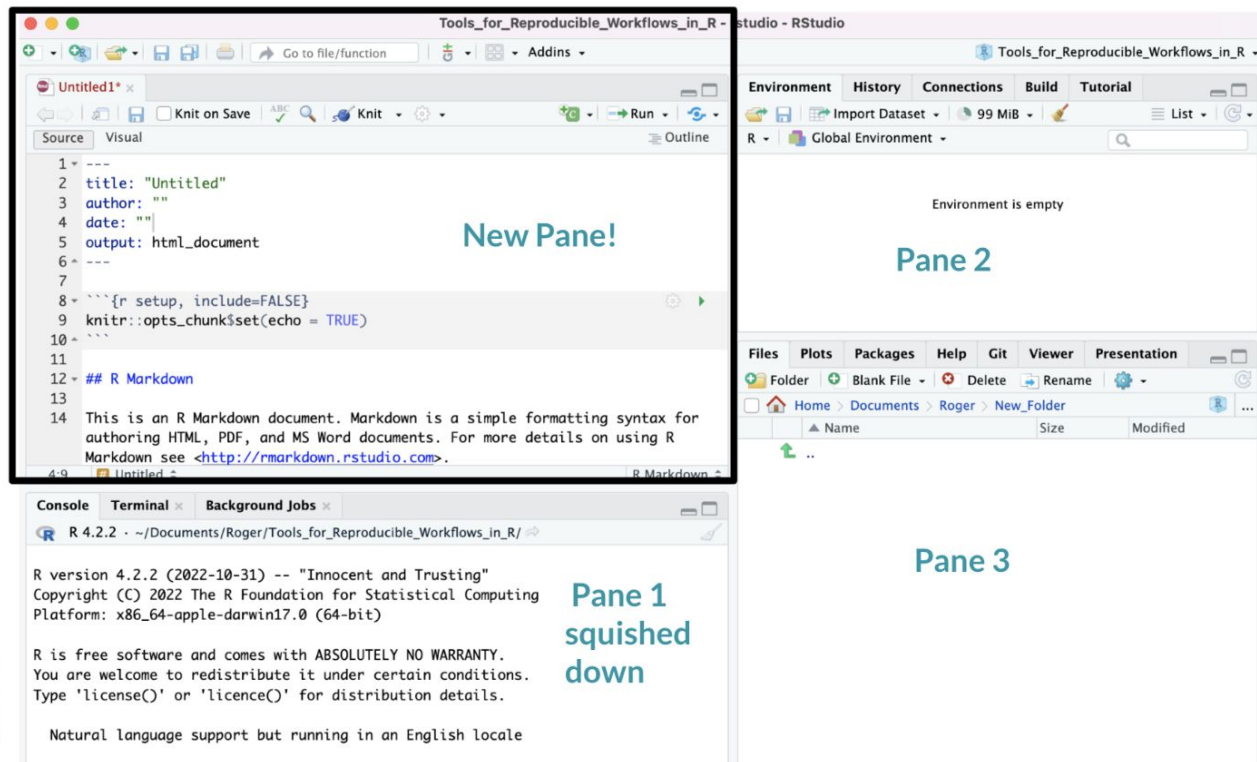
## Hidden Pane

To save a copy of your code. You must open a file first - this will open a 4th pane. These files include Scripts or what are called R Markdown files.



# Hidden Pane

Nice! now we have a place to save code! This is where we will mostly be working.



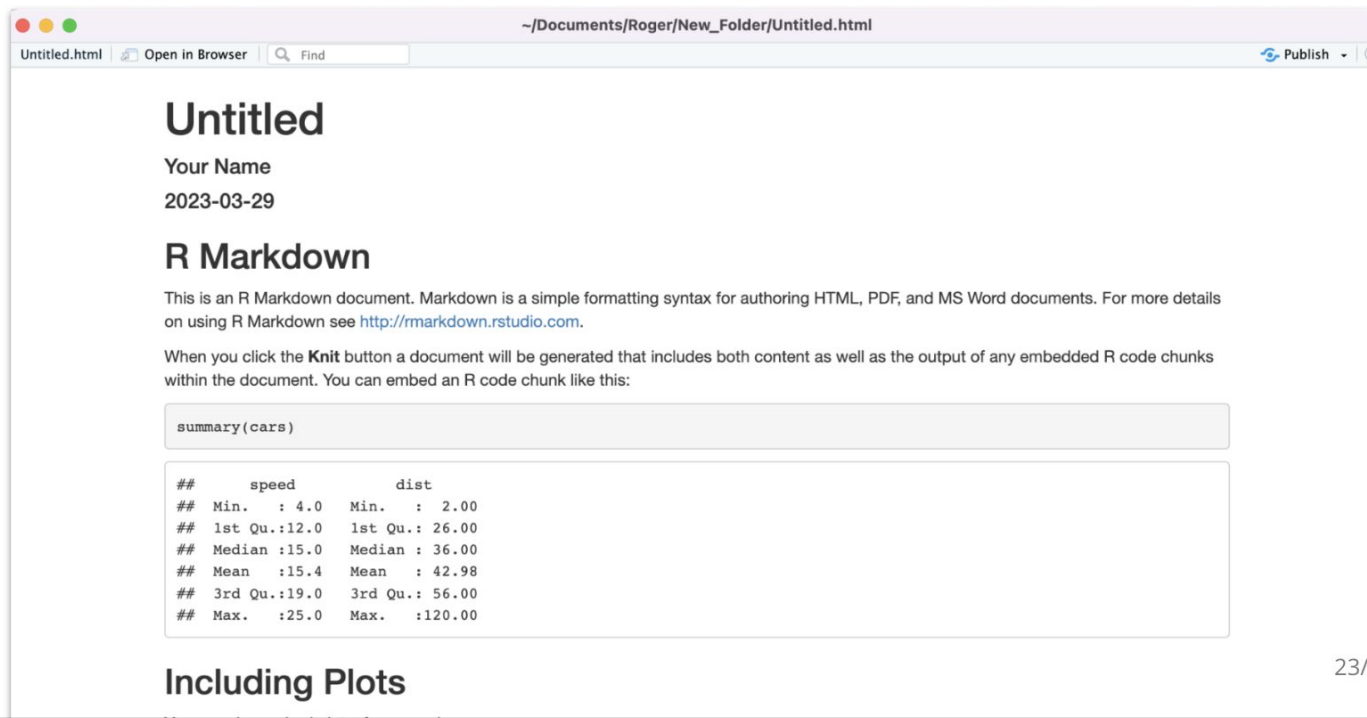
# R Markdown file

R Markdown files (.Rmd) help generate reports that include your code and output.

1. Helps you describe your code
2. Allows you to check the output
3. Can create many different file types

# Nice report!

This generates a nice report that you can share with others who can open in any browser.



The screenshot shows a web browser window displaying an R Markdown report. The browser's address bar shows the file path: `~/Documents/Roger/New_Folder/Untitled.html`. The report content includes a title, a date, a section header, an introductory paragraph about R Markdown, a code chunk for `summary(cars)`, and its corresponding output table.

## Untitled

Your Name  
2023-03-29

## R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

##	speed	dist
##	Min. : 4.0	Min. : 2.00
##	1st Qu.:12.0	1st Qu.: 26.00
##	Median :15.0	Median : 36.00
##	Mean :15.4	Mean : 42.98
##	3rd Qu.:19.0	3rd Qu.: 56.00
##	Max. :25.0	Max. :120.00

## Including Plots

# Summary

- Repeatable → Reproducible → Replicable
- Others know what you did well enough to use your data and code and get the same results
- Reproducibility:
  - Helps make science more efficient!
  - Helps your future self and others know what you did
    - Saving time and effort
  - Makes it easier to adjust or update your work
- Reproducible does not mean correct! But it is a good first step to help identify if your work is correct
- Organizing files, naming files consistently and informatively helps! R Markdown files do too - more about that next!